# Name: *ANSWERS*

## ENERGY, WORK AND POWER

**W = Fd** **P = $\frac{E}{t}$** **$E_p$ = mgh** **$E_k$ = ½$mv^2$**

1. Complete the following chart.

| Quantity | Symbol | Unit | Symbol |
|---|---|---|---|
| Work | ***W*** | ***Joule*** | ***J*** |
| ***Power*** | P | ***Watt*** | ***W*** |
| ***Energy*** | ***E*** | Joule | ***J*** |

2. Write a clear definition of Work. ***Work is the energy transferred from one form to another***

3. Write a clear definition of Power. ***Power is the rate of doing work***

4. 400J of electrical energy in a battery is turned into light energy. How much work has been done by the battery? ***400J***

5. A boy lifts a heavy book up from the floor and puts it on the table. The book now has 15J of energy. How much work has the boy done on the book? ***15J***

6. The boy picks up another book from the floor which has a mass of 1.2kg. He lifts it 90cm and puts it on the table top.
   (a) How much gravitational potential energy has the book gained? (Use g = 10 $Nkg^{-1}$)
   ***E = mgh = 1.2x10x0.9 = 10.8 J***
   (b) How much work has the boy done in lifting the book?
   ***Work done = energy transferred = 10.8 J***

7. A car with mass of 1200 kg accelerates from rest to reach a speed of 20 $ms^{-1}$.
   (a) How much kinetic energy does the car now have?
   ***Ek = 0.5x1200x20x20 = 240,000 J***
   (b) How much work did the motor do in accelerating the car?
   ***Work done = energy transferred = 240,000 J***
   (c) If the car took 6 seconds to accelerate to 20$ms^{-1}$, what was the power output of the engine?
   ***Power = E/t = 240000/6 = 40,000 W***

8. An electric light bulb is rated at 100W. If the light is left on for ten minutes, calculate the energy it uses.

***E = Pxt = 100 x 10 x 60 = 60,000 J***

9. An electric jug is rated at 1.8 kW. How many Watts is this? ***1800 W***

If this jug takes 2 minutes to boil some water, how many Joules of heat energy did it produce?

***E = Pxt = 1800 x 2 x 60 = 216,000 J***

10. When John worked in his holidays he had to wheelbarrow cement along a track from a concrete truck to a building site. He reckoned that he was using about 200N of force to push the wheel barrow and he had to push it a distance of 30m.

Calculate the work he did every time he delivered a barrow of cement.

***W = Fxd = 200 x 30 = 6000 J***

11. Complete the following chart. Remember, W = Fd

| **Work (J)** | ***36*** | ***50*** | 2000 | 4.8 | 55 | 330 | 6 | 0.5 |
|---|---|---|---|---|---|---|---|---|
| **Force (N)** | 12 | 100 | ***40*** | ***0.08*** | ***5*** | 15 | 30 | 1.7 |
| **Distance (m)** | 3 | 0.5 | 50 | 60 | 11 | ***22*** | ***0.2*** | ***0.3*** |

12. Fan A uses 200J of electrical energy every 10 seconds. What is its power rating?

***20 W***

Fan B is twice as "powerful" as fan A. How much energy will it use every second?

***40 J***

13. A boy takes 5 seconds to pull a fish up into the boat. The average force he uses while pulling it in is 30N and he has to reel in 20m of line.

(a) What work has the boy done? ***W = 30 x 20 = 600 J***

(b) What power has he used in the process? ***P = 600/5 = 120 W***

14. Complete the following table. Remember $P = \frac{E}{t}$

| **Power (W)** | ***12*** | ***0.13*** | ***8.3*** | 75 | 700 | 1kW | 2kW | 25 |
|---|---|---|---|---|---|---|---|---|
| **Work (J)** | 60 | 32 | 1kJ | ***4125*** | ***8400*** | 100 | 9kJ | 200 |
| **Time (s)** | 5 | 240 | 2min | 55 | 12 | ***0.1*** | ***4.5*** | ***8*** |